fastMRI: Attentive Reconstruction

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Abstract

Magnetic Resonance Imaging (MRI) scanning, which provides unparalleled insights into the intricate structures and functions of the human body, has long been recognized as a crucial and invaluable tool for diagnosing various medical conditions. However, its long acquisition time brings discomfort and high examination cost to patients. In this project, a deep-learning based MRI reconstruction is conducted on under-sampled single-coil knee data to shorten the acquisition time as well as achieving high resolution images. A baseline U-net model is trained using Structural Similarity Index Metrics (SSIM) as loss function. Besides, a U-net architecture is also trained with the metrics of combination loss of MSE and SSIM to see how this combination would affect the model performance. In addition, hyper-parameters including dropout probability and SSIM kernel size are also tested and analyzed. MRI reconstruction is performed on our pre-trained models with different metrics and also compared to ground truth images.

1. Introduction

Magnetic Resonance Imaging (MRI) is one of the most powerful imaging techniques for medical image diagnosis due to its non-invasive, non-intrusive nature and excellent soft tissue contrast for internal structures such as organs and muscles, which don't show up on x-rays in a useful way. However, long acquisition time in MRI, which ranging from 15 to 90 minutes, leads to low patient throughput, discomfort and high exam cost. As a consequence, speeding up MRI acquisition time while maintaining its high imaging quality has been an popular research topic over the past few decades.

In order to achieve high-quality image with better spatial resolution, spatial frequency sampling is necessary and crucial for capturing detailed information to better represent various features and structures. The object's spatial frequency representation – the Fourier space known as k-space – must be fully sampled in order to avoid Shannon-Nyquist aliasing, which limits the MR imaging speed due to the large sampling number.

The two most common techniques developed to maintain the imaging quality while reducing the sampling number are Parallel Imaging and Compressed Sensing. Parallel Imaging is a technique developed to speed up the acquisition by using multiple receiver coils to exploit the spatial sensitivity information and reduce the number of phase-encoding steps required for reconstruction [18]. SENSE (SENSitivity Encoding) is one of the most common parallel imaging technique, which utilizes sensitivity profiles of multiple receiver coils [13]. GRAPPA, or GeneRalized Autocalibrating Partially Parallel Acquisitions, estimates missing k-space data points by using acquired auto-calibration signals (ACS) combined with local known data [6]. Other techniques including ARC (Autocalibrating Reconstruction for Cartesian imaging) extends the principal of GRAPPA by using a 3D kernel to synthesize missing data and neighboring known data from all three directions [1]; TGRAPPA, or temporal GRAPPA, allows one to easily update the coil weights dynamically by combining a time-interleaved acquisition scheme with conventional GRAPPA [2]. However, the acceleration factor is limited by the number and arrangement of receiver coils, which potentially introduces some imaging artifacts [4].

On the other hand, Compressed Sensing based MRI (CS-MRI) bypasses the Nyquist-Shannon sampling criteria that only a small subset of k-space data is required to generate full reconstruction via a computational method [16]. Three fundamental requirements for application of CS to MRI reconstruction including transform sparsity, seudo-random undersampling, and non-linear reconstruction [5]. Theoretically, CS-MRI can achieve a reconstruction without deterioration of image quality by performing non-linear optimization on randomly undersampled raw data, assuming a proper sparsifying transformation such as wavelet transformation is found. Though studies have shown that Compressed Sensing successfully accelerate MRI, studies involving greater number of radiologists and images must be completed.

Furthermore, approaches of Parallel Imaging combined with Compressed Sensing is developed to further shorten acquisition time. Attempts to combine these two tech-

Random order. All authors made equal contribution.

niques have mostly focused on extensions of iterative SENSE [12]with SparseMRI [9]. More recently, optimized iterative method, SPIRiT and its CS-combined extention, l1-SPIRiT are presented by Murphy et al [11].

More recently, deep neural networks began outperforming other machine learning models and specifically have a huge potential for medical imaging technology and medical diagnostics. Given the enormous amount of data containing extremely valuable signals and information generated by healthcare providers, deep learning methods are expected to learn more natural and high-level abstract underlying features instead of handcrafted feature extraction using traditional methods.

A deep architecture ADMM-Net is presented by Yang et al. in 2016, that the deep networks is derived from the ADMM algorithm for optimizing a CS-based MRI model [17]. Qin et al. proposed a convolutional recurrent neural network architecture for dynamic MRI reconstruction [14]. Knoll et al. also explored the potential for transfer learning and the generalization of learned image reconstruction [8]. Chen et al. presented an approach to improve reconstruction speed by using a variational network(Varnet) and achieved an improved overall image quality [3]. The Varnet is also one of the commonly used architectures in fastMRI, a collaborative research project from Facebook AI Research (FAIR) and NYU Langone Health, which is also the project we are working on.

In this project, a convolutional neural network U-Net is selected as our baseline model to be trained for MRI reconstruction with undersampled k-space data. The training process is performed on the entire knee single-coil dataset from NYU FastMRI dataset, including the single-coil training dataset and single-coil validation dataset. The dataset contains two types of data we care about, including raw k-space data and ground truth images. These two types of data will be wrapped as iterable datasets to enable easy access to data for both training and testing phases.

2. Approach

2.1. MRI K-space and Image Reconstruction

In the FastMRI dataset, k-space data serves as the matrix representation of raw signal values obtained during the MRI process.[10]

Each element of the k-space matrix corresponds to an echo of the signal, uniquely encoding both phase and frequency information. The total number of elements in the matrix reflects the number of samples acquired during the MRI procedure. The vertical axis of the matrix represents variations in phase, while the horizontal axis represents variations in frequency.

K-space images themselves are not directly interpretable as conventional images providing insight into a patient's condition. Therefore, image processing is necessary to translate information from signal space to spatial space. Given that k-space encoding involves Fourier transformation for specific linear phases, inverse Fourier transformation is utilized to reconstruct the images. There are several critical relationships between k-space images and their corresponding reconstructions[7]:

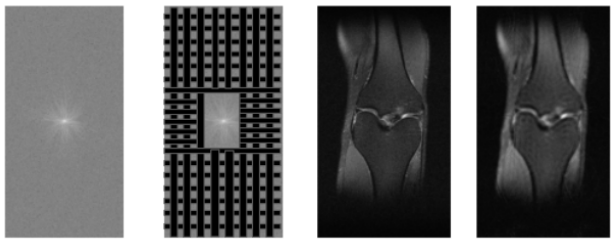


Figure 1. The images from the left to the right are k-space image, sub-sampling on k-space image, reconstruction image from fully-sampled k-space, reconstruction image from sub-sampled k-space.

1. Reconstructed images maintain the same dimensions as their respective k-space images.
2. Image resolution is influenced by the field of view (FOV) and the matrix size; more samples yield greater information and higher resolution.
3. The periphery of k-space preserves margin and edge details of reconstructed images. Removing this outer region results in blurred images with obscured margins.
4. The center of k-space contains crucial contrast information for the reconstructed images. Eliminating this central portion yields images with preserved margins but reduced contrast.

MRI scans may require hours to collect sufficient data for fully sampled k-space. To expedite MRI completion time and enhance diagnostic efficiency, image reconstruction techniques coupled with image refinement algorithms are essential. In our experiment, we employ the U-net deep neural network to refine images. Sub-sampled k-space data is obtained by applying a mask to the original k-space images, where the mask is generated based on acceleration rate and window size. Acceleration rate determines the extent of sampling reduction, thus shortening MRI duration, while the window size dictates the level of blurring in the input images.

2.2. U-Net Deep Neural Network

2.2.1 U-net Structure

The U-Net architecture comprises two interconnected paths: the contracting path and the expansion path. [15]As

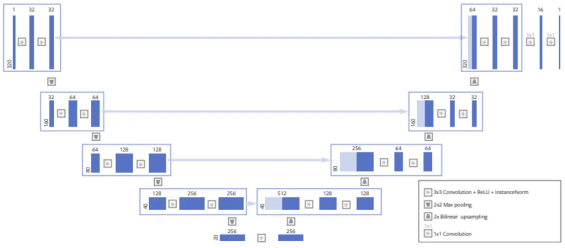


Figure 2. Unet Structure

shown in Figure 2, the contracting path begins with two successive layers of 3x3 convolutions, followed by rectified linear units (ReLU), and culminates in a 2x2 max pooling operation. This path facilitates downsampling of the input tensor, progressively extracting features from the images.

Conversely, the expansion path involves two additional layers of 3x3 convolutions, which amplify the number of feature channels through upsampling. Like the contracting path, it incorporates ReLU activations and upsampling layers. As the network progresses upwards, intermediate features are upsampled to generate the final output images.

At each depth of the U-Net, the left side of the contracting path is directly linked to the right side of the expansion path. This ensures that corresponding features from the contracting path are concatenated with the upsampling feature channels. This fusion of context and semantic information enhances the network's ability to perform tasks such as segmentation or image refinement. In our experiment, we refer the U-net model implementation by Meta, the code is accessed from (Github Link)

2.2.2 Why U-net?

The U-Net model was initially introduced for biomedical image segmentation tasks, which instills confidence in utilizing it for training MRI datasets. Besides,U-Net facilitates the mapping of input image pixels to corresponding output pixels.

Moreover,the U-Net model can achieve precise segmentation even when trained on a small number of training images. This is particularly advantageous when dealing with limited access to MRI images and constrained GPU resources for training medical image refinement models.

In addition, the U-Net's U-shaped architecture, allows the network to capture ample context and semantic information, makes it possible for learning the transition from blurred input images to their target counterparts

2.3. Loss Determination

Our experiment aims to assess the performance of U-Net models in the context of image refinement training using various loss functions. Specifically, we trained separate models with Mean Square Error (MSE) and Structural Similarity Index (SSIM) loss functions. Additionally, we developed another model that combines the losses from both SSIM and MSE, allowing for tunable weights for each loss.

2.3.1 MSE Loss

The Mean Squared Error (MSE) is calculated as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

where n is the number of samples, y_i represents the actual value of the ith sample, and $\hat{y}_i$ represents the predicted value of the ith sample.

In our research, the selection of MSE can reflect the overall disparities in pixel values between the target and input images.

2.3.2 SSIM Loss

The Structural Similarity Index (SSIM) combines three comparison functions: Luminance, Contrast, and Structure. It is calculated as:

$$\text{SSIM}(x, y) = \text{Luminance}(x, y) \cdot \text{Contrast}(x, y) \cdot \text{Structure}(x, y) \tag{2}$$

where:

$$\text{Luminance}(x, y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1} \tag{3}$$

$$\text{Contrast}(x, y) = \frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2} \tag{4}$$

$$\text{Structure}(x, y) = \frac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3} \tag{5}$$

In these equations, x and y represent the two input images, μ_x and μ_y are their average pixel values, σ_x and σ_y are their variances, and σ_{xy} is their covariance. Constants c_1, c_2, and c_3 are used to stabilize the division with weak denominators.

Unlike MSE, SSIM better captures the local structural changes between input and target images.

2.3.3 Combination Usage of MSE loss and SSIM Loss

The combined loss, denoted as combine_loss, is calculated as follows:

$$\text{combine_loss} = \alpha \times \text{MSE} + \beta \times \text{SSIM_LOSS} \tag{6}$$

where α represents the contribution of the Mean Squared Error (MSE), and β represents the contribution of the Structural Similarity Index (SSIM) loss.

3. Experiments and Results

3.1. Data

For our experiments, several data sets were evaluated based on the MRI image reconstruction purpose.The followings are the primary sources for the consideration:

FastMRI data sets by Facebook AI Research and NYU Langone Health: This data set is one of the largest available for MRI image reconstruction, encompassing both knee and brain imaging. The fastMRI data sets offer a comprehensive collection of raw k-space data.

MRNet data sets by Stanford ML Group: Provided by the Stanford ML Group, this collection includes annotated MRI scans of the knee, which are useful for training models to recognize specific areas of pathology.

Considering the limited timeframe of our project and the extensive computational resources required, the single-coil fastMRI data sets was our primary data source. This decision was influenced by the size and diversity of the fastMRI data, as well as its accessibility and relevance to our project goals.

3.2. Data Processing

In order to effectively handle and transform the raw k-space data to a format suitable for the DNN training, the data processing pipeline for the data set is design as shown in this pipeline:

3.2.1 Data Ingestion

File Retrieval: The pipeline was implemented by retrieving a specified number of HDF5 (.h5) files from a directory, with a maximum limit to manage computational load. Each file contains k-space data, target reconstruction images, and the associated file name.

3.2.2 Data Preprocessing

Reading and Filtering Data: The HDF5 files were processed to extract slices of k-space and target images. This selective retrieval is crucial for focusing on relevant parts of the data and reduce the training time to increase training/tuning efficiency.

Image Reconstruction: From the k-space data, an inverse Fast Fourier Transform (IFFT) was performed to reconstruct the spatial MRI images. This step is pivotal in transforming the frequency-domain data into a visual format that can be used for further processing and analysis.

Image Cropping and Resizing: To enhance the model's focus on diagnostically relevant areas, the reconstructed images are cropped and resized to avoid the dark border areas. This step ensures that the neural network training concentrates on the most informative parts of the images.

Data Loading: Data loaders were employed in order to batch the data into manageable sizes for processing by the DNN. This batching is crucial for efficient memory management and computational performance during model training.

3.3. Model Training

3.3.1 Initialization and Configuration:

Model Configuration: The Unet model was configured with parameters specified in our configuration file/arguments, including acceleration rate, drop rate, epochs, learninig rate, SSIM window size and other parameters.

Optimizer: The Adam optimizer was leveraged for its efficiency in handling sparse gradients on noisy problems, with a learning rate defined in the configuration file.

Loss Metrics: Depending on the experiments, mean squared error (MSE) and structural similarity index (SSIM) were used as the loss metric. SSIM is particularly useful for preserving the visual quality of images.

3.3.2 Epoch Execution

Epoch Execution: Training was performed over several epochs (iterations over the entire dataset), as defined in the configuration. Each epoch involved processing all batches of training data, followed by validation with a separate set of data to monitor model performance and prevent overfitting.

Loss Computation and Backpropagation: For each batch, the model outputs were computed, followed by loss calculation. The gradients of the loss were then backpropagated through the network to update the model weights.

3.3.3 Monitoring and Adjustment

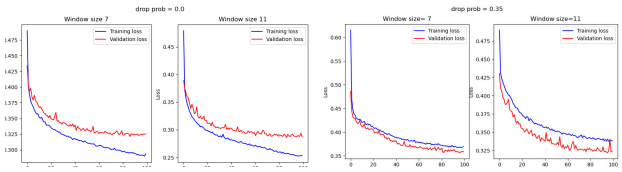
Performance Tracking: Throughout the training process, the loss was recorded at the end of each epoch for both training and validation phases to monitor the model's performance and convergence. **Model Saving:** After each epoch, the model state was saved. This allows for recovery and further analysis, ensuring that the best-performing model is retained. **Testing and Prediction:** Following training, the model can be loaded and used to make predictions on new data. The outputs were visualized to assess the quality of reconstruction and compare against ground truth images.

3.3.4 Hyper Parameters Tuning

The major hyper parameters we are tuning during the training are listed in Table1

Parameter	Description
Batch Size	How many images are trained per batch
Slices	Which slices we are training on
Accelerate Rate	How many times faster we want to achieve
Dropout Probability	Percentage of parameters dropped per iteration
Window Size	Moving window size when calculating SSIM loss

Table 1. Hyperparameter Settings



(a) (b)

Figure 3. (a)Loss vs Epoch number under different window size (dropout probability=0.0) (b)Loss vs Epoch number under different window size (dropout probability=0.35)

(a) (b) (c)

Figure 4. (a) Reconstruction via zero-filled IFFT (b) Reconstruction with Unet model trained with SSIM loss (c) Reconstruction from fully-sampled k-space data

3.3.5 Visualization and Results

In the results section, we mainly present the two models archived the top performances. with Post training, we plot the training and validation loss over each epoch to visualize the learning progress and identify any potential issues such as overfitting.

3.4. Results

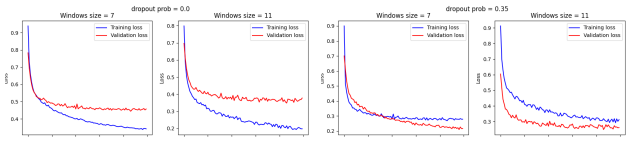
3.4.1 Model Trained with SSIM

In this project, different kernel size of 7 and 11 for SSIM and dropout probability of 0.0 and 0.35 are selected for analyzing how these parameters will affect the model performance. The learning curves regarding training and validation loss are presented in Figure 3. As you can see in Figure 3 (a), learning curves shows a signal of over-fitting, since training curves keep decreasing while validation loss keeps constant. On the contrary, the models trained with dropout probability of 0.35, though both have a larger validation loss at epoch 100, the gap between the training and validation curves is much smaller. In addition, both training curves lie above the corresponding validation curves indicates that to different degrees the possible under-fitting of the models. Better dropout probability could possibly lie between 0.25 to 0.3 to prevent model from over-fitting as well as gaining sufficient complexity. As for window size chosen for SSIM loss, models with window size 11 have better performance that corresponding losses are more efficiently reduced. It could be the reason that smaller window size could lead to noisier estimates due to lack of samples.

The reconstruction using baseline Unet model trained with SSIM loss is presented in Figure 4. The reconstruction image is generated with the best model trained with slice number 19, 20, 21, and SSIM as loss type for optimization. Compared to the image generated by directly applying inverse Fourier Transform to under-sampled data shown in Figure 4 (a), the image with deep learning reconstruction has clearer, smoother boundary and provides more accurate information. Compared to the ground truth image reconstructed from full-sampled data, however, details such as the soft tissue boundary on the upper right part for deep learning reconstruction is not accurate enough for medical diagnosis. This could be the reason that model is trained with limited slices, which leads to lack of information and failure of extracting higher level abstract features.

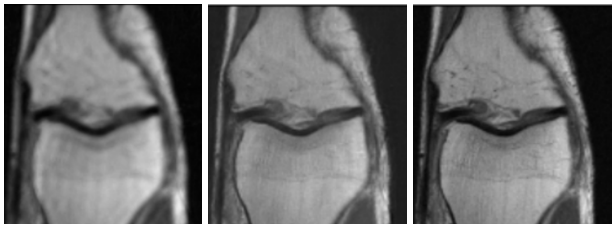
3.4.2 Model Trained with Weighted Sum of MSE and SSIM

Training with weighted sum of MSE and SSIM as loss function was proved as a valid loss choice training the U-Net model as shown in 5. The output construction is clear with context and include more details as shown in6. There are several key findings associated with the training results. Data preprocessing is vital for effective network learning. After applying inverse FFT to process k-space data, resulting complex numbers tend to be small, hindering effective learning. Hence, normalization post-inverse FFT is essential. Initially, we attempted Min-max normalization but found extreme outliers in some images, skewing data distribution. To mitigate this, mean normalization and clipping extreme values were employed. This led to significant improvement, enabling the model to effectively learn and correct subsampled k-space images based on fully sampled target images. MSE ensures accurate pixel-level reconstruction but may produce overly smooth results. Combining it with SSIM leverages both fidelity and perceptual quality. During training, we observed fluctuations between MSE and SSIM loss. Assigning different weights to each loss function helps balance the training process and optimize image quality.



(a) (b)

Figure 5. Loss vs number of Epoch under different window size when training with weighted sum of mse and ssim (a)dropout probability=0.0 (b) dropout probability=0.35



(a) (b) (c)

Figure 6. Detail comparisons between (a) Reconstruction via zero-filled IFFT (b) Output reconstruction image of model trained with weighted sum of MSE and SSIM (c) Ground truth image

4. Limitations and Future Work

One significant limitation is the constrained access to GPU/CPU resources, which restricts the exploration of various hyperparameter combinations, such as larger batch sizes. This limitation indirectly impedes our ability to optimize model performance. In the subsequent section, we delve into discussions regarding cloud ML infrastructure as a potential solution.

Another limitation is that our models were exclusively trained on the single-coil dataset, with experimentation limited to a subset of slices. Training on a multi-coil dataset holds promise for improving image reconstruction, particularly for knee images.

Exploring alternative loss functions beyond MSE or SSIM is crucial for enhancing our understanding of image semantics. Future work could involve investigating loss functions better suited to capturing semantic information.

Furthermore, data augmentation played a vital role in the success of U-Net. However, we were unable to fully explore certain augmentation methods that may further enhance refinement results.

4.1. System Dataflow and Cloud Implementation

The Figure 7 shows the training system dataflow which includes Data Ingestion and Preprocessing, Hyperparameters Configuration, and Training and Validation Process. The model training is an iterative process, where the loss graphs guide potential adjustments in hyperparameters or model architecture. With each iteration, the aim is to minimize the loss and enhance the model's predictive accuracy on the validation dataset, which is indicative of its expected performance on real-world MRI data.

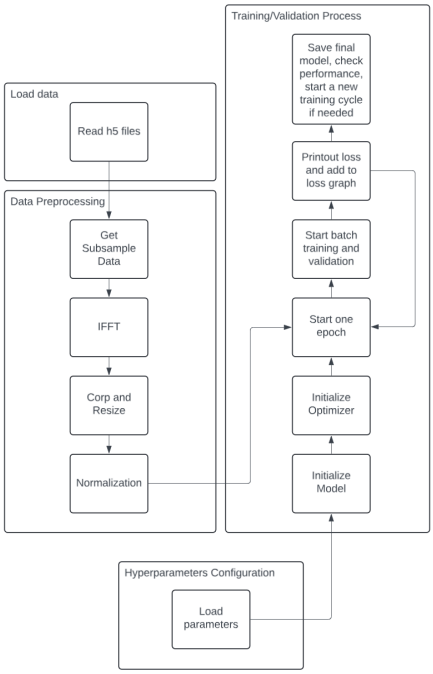


Figure 7. Training Pipeline Dataflow

In a real-life scenario, Machine Learning infrastructure is a complex but critical part of developing, deploying, and maintaining AI systems. The process starts with raw data that goes through a feature pipeline, consisting of feature engineering, to produce a feature store for continual training. This data feeds into the model training phase, after which the model is registered and deployed for inference, available for both online direct inference and batch scoring. Post-deployment operations include A/B testing, model performance management, and model observability. Additionally, there's an interface for a web application that likely utilizes the model for practical applications.

5. Work Division

Student Name	Contributed Aspects and Details
Chuying Lu	Built basic framework, implemented parts of the data processing features, trained and tuned the model. Contributed to the reports.
Fang Zou	Implemented part of data transform feature. Contributed to the reports.
Tianquan Guo	Implemented Parts of the data preprocessing features and model training codes. Contributed to the reports.
Yawen Xiao	Built basic framework,implemented parts of the models training codes, trained and tuned the model. Contributed to the reports.

Table 2. Contributions of team members.

References

[1] R Bammer, E Rubesova, A White, ST Skare, P Beatty, and A Brau. Improving image quality in fetal mri using autocalibrating reconstruction for cartesian (arc) sampling. In *Proc. Intl. Soc. Mag. Reson. Med*, volume 14, page 2318, 2006.

[2] Felix A Breuer, Peter Kellman, Mark A Griswold, and Peter M Jakob. Dynamic autocalibrated parallel imaging using temporal grappa (tgrappa). *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 53(4):981–985, 2005.

[3] Feiyu Chen, Valentina Taviani, Itzik Malkiel, Joseph Y Cheng, Jonathan I Tamir, Jamil Shaikh, Stephanie T Chang, Christopher J Hardy, John M Pauly, and Shreyas S Vasanawala. Variable-density single-shot fast spin-echo mri with deep learning reconstruction by using variational networks. *Radiology*, 289(2):366–373, 2018.

[4] Anagha Deshmane, Vikas Gulani, Mark A Griswold, and Nicole Seiberlich. Parallel mr imaging. *Journal of Magnetic Resonance Imaging*, 36(1):55–72, 2012.

[5] Sairam Geethanath, Rashmi Reddy, Amaresha Shridhar Konar, Shaikh Imam, Rajagopalan Sundaresan, Ramesh Babu DR, and Ramesh Venkatesan. Compressed sensing mri: a review. *Critical Reviews™ in Biomedical Engineering*, 41(3), 2013.

[6] Mark A Griswold, Peter M Jakob, Robin M Heidemann, Mathias Nittka, Vladimir Jellus, Jianmin Wang, Berthold Kiefer, and Axel Haase. Generalized autocalibrating partially parallel acquisitions (grappa). *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 47(6):1202–1210, 2002.

[7] MD Joseph W. Owen. Mr physics 4 - k space. YouTube video, 2020.

[8] Florian Knoll, Kerstin Hammernik, Erich Kobler, Thomas Pock, Michael P Recht, and Daniel K Sodickson. Assessment of the generalization of learned image reconstruction and the potential for transfer learning. *Magnetic resonance in medicine*, 81(1):116–128, 2019.

[9] Michael Lustig, David Donoho, and John M Pauly. Sparse mri: The application of compressed sensing for rapid mr imaging. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 58(6):1182–1195, 2007.

[10] D. Moratal, A. Vallés-Luch, L. Martí-Bonmatí, and M. E. Brummer. k-space tutorial: An mri educational tool for a better understanding of k-space. *Biomed Imaging Interv J*, 2008.

[11] Mark Murphy and Miki Lustig. Fast l1-spirit compressed sensing parallel imaging mri: Scalable parallel implementation and clinically feasible runtime.

[12] Klaas P Pruessmann, Markus Weiger, Peter Börnert, and Peter Boesiger. Advances in sensitivity encoding with arbitrary k-space trajectories. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 46(4):638–651, 2001.

[13] Klaas P Pruessmann, Markus Weiger, Markus B Scheidegger, and Peter Boesiger. Sense: sensitivity encoding for fast mri. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 42(5):952–962, 1999.

[14] Chen Qin, Jo Schlemper, Jose Caballero, Anthony N Price, Joseph V Hajnal, and Daniel Rueckert. Convolutional recurrent neural networks for dynamic mr image reconstruction. *IEEE transactions on medical imaging*, 38(1):280–290, 2018.

[15] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, volume 9351, pages 234–241, 2015.

[16] Guang Yang, Simiao Yu, Hao Dong, Greg Slabaugh, Pier Luigi Dragotti, Xujiong Ye, Fangde Liu, Simon Arridge, Jennifer Keegan, Yike Guo, et al. Dagan: deep de-aliasing generative adversarial networks for fast compressed sensing mri reconstruction. *IEEE transactions on medical imaging*, 37(6):1310–1321, 2017.

[17] Yan Yang, Jian Sun, Huibin Li, and Zongben Xu. Deep admm-net for compressive sensing mri. In *Proceedings of the 30th international conference on neural information processing systems*, pages 10–18, 2016.

[18] Jure Zbontar, Florian Knoll, Anuroop Sriram, Tullie Murrell, Zhengnan Huang, Matthew J Muckley, Aaron Defazio, Ruben Stern, Patricia Johnson, Mary Bruno, et al. fastmri: An open dataset and benchmarks for accelerated mri. *arXiv preprint arXiv:1811.08839*, 2018.